

see-saw: A Kinetic Installation that Unfolds in Silence Driven by Liquid Flow

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ABSTRACT

see-saw is a kinetic installation unfolding silently, driven by the behavior of liquid flowing inside the objects. A group of these objects, shaped like seesaws, are aligned. By changing the position of the center of gravity of the liquid inside, each see-saw alters its inclination dynamically. The behavior of liquid is realized using the Electrohydrodynamics (EHD) phenomenon. We generate these movements that appear to depart from conventional physics by electrically manipulating an invisible driving force to move the liquid against gravity. This installation embodies a natural and smooth kinetic aesthetic that blends seamlessly into the surrounding environment through silent movement.

CCS CONCEPTS

• **Applied computing** → Media arts.

KEYWORDS

Programmable Materials, Hydraulic, Kinetic Installation, Electrohydrodynamics

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1 INTRODUCTION

The primary issue of our artwork lies in the overlooked beauty of natural phenomena, which often go unnoticed in our everyday lives unless we pay close attention. Thus, the concept of this work is to offer the audience an experience that allows them to re-appreciate this beauty. One example of this beauty can be seen in the way leaves on trees sway sporadically when a breeze blows through. Countless variables, such as the direction and strength of the wind, interact to create beautiful, accidental movements. These movements never repeat identically and vary depending on the place and time of observation. Although this is just one example, such occurrences are ubiquitous in our daily environments. Our artwork is motivated by a desire not to discard such uncontrollable randomness as noise, but instead to respect and utilize its behaviors based

on their unique characteristics and mechanisms. We further explore the possibilities of technologies that can create new value, while leaving some room for interpretative flexibility in the artwork.

Under such motivations, we turned our attention to liquids. Liquids are highly familiar to us as they are frequently encountered in daily life. Furthermore, the formless nature of liquids allows expressions that were difficult to achieve with mechanical structures like motors, such as dynamic shape changes, soft tactile sensations, and supple movements. Additionally, by utilizing thermal conductivity for temperature control [Han et al., 2018; Liu et al., 2021], color for visual expressions [Goldman and Manganiello, 2020; Inoue et al., 2018; Mor et al., 2020; Popp, 2011], and chemical reactions for specific effects [Lu et al., 2021; Wehner et al., 2016], the unique properties of liquids offer a wide range of expressive possibilities. Several artists and researchers have utilized liquids in their artwork and interface research thus far. However, most of them use off-the-shelf hydraulic pumps to drive the liquid, and the loud noise generated during operation is a common issue. Even if they manage to create soft movements, the overall system often gives an artificial impression, making it difficult to blend into the surrounding environment. As a result, the audience may feel a sense of discomfort when it operates.

Therefore, we propose "see-saw," a kinetic installation that silently unfolds movements through the behavior of the liquid flowing inside the object, as shown in Figure 1. This design ensures a soft movement while also aiming to integrate into the surrounding environment seamlessly. These objects, shaped like see-saws, are arranged at random angles and intervals without any regularity. By changing the position of the center of gravity of the liquid inside, each see-saw alters its inclination dynamically. The behavior of the liquid is realized using a module that applies the Electrohydrodynamics (EHD) phenomenon [Cacucciolo et al., 2019; Pickard, 1963a; Pickard, 1963b; Richter and Sandmaier, 1990]. This results in a silent movement without any actuator noise. One of the etymologies of the word "seesaw" is that it is a repetition of "see" and "saw", hence the name "seesaw". We generate these movements that appear to depart from conventional physics by electrically manipulating an invisible driving force to move the liquid against gravity. This installation embodies a natural, smooth kinetic aesthetic that blends into the surroundings through silent movement. Through this work, we aim to provide the audience with an opportunity to re-appreciate the unnoticed beauty of natural phenomena in everyday life.

2 PRINCIPLE OF THE BEHAVIOR OF LIQUID UTILIZING EHD PHENOMENON

In our proposal, we utilize the Electrohydrodynamics (EHD) phenomenon as a behavior of the liquid, employing an EHD pump. The EHD phenomenon is the flow phenomenon of a dielectric liquid in

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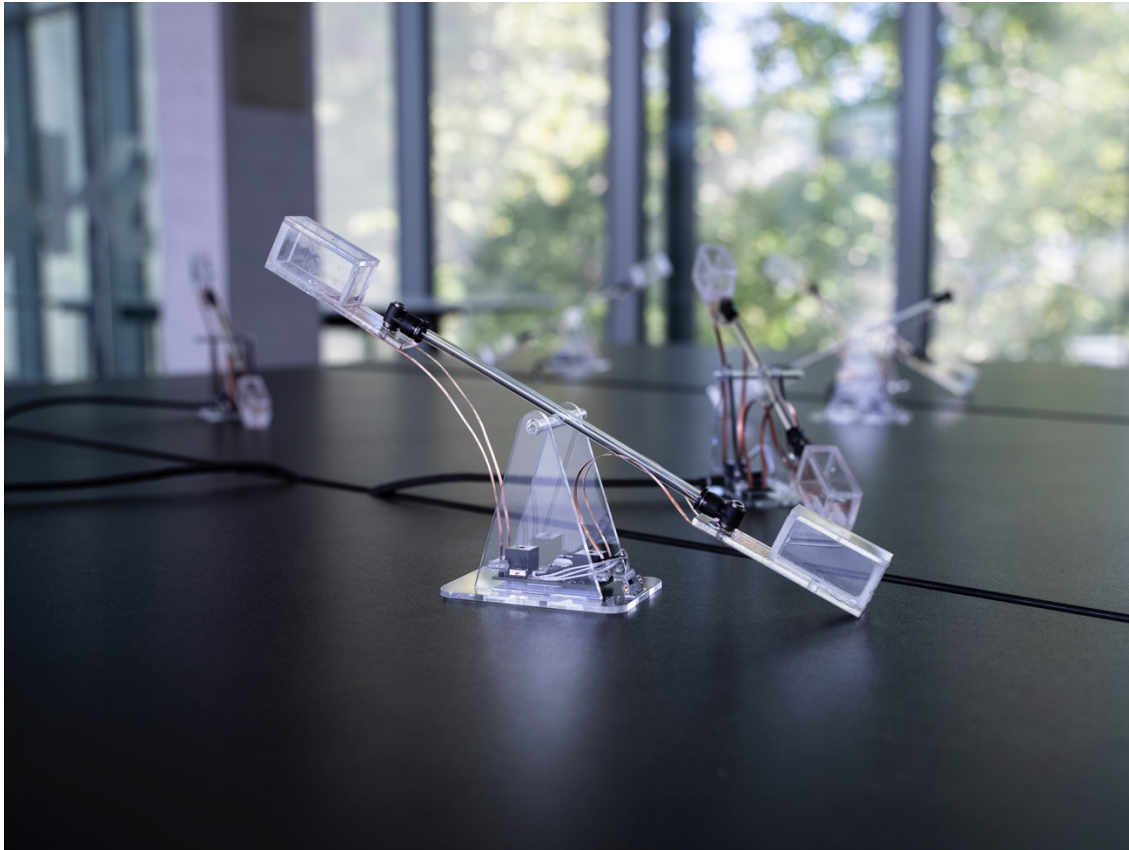


Figure 1: "see-saw," a kinetic installation, operates silently based on the changes in the center of gravity caused by the behavior of liquid flow inside the objects. (©Takafumi Morita. Photo: Yasuaki Kakehi.)

a high electric field induced by electrochemical reactions. Figure 2 shows the mechanism of the EHD phenomenon. When high voltage is applied to a comb-shaped electrode, a high electric field is generated at each tooth of the comb. The liquid flow in the direction opposite to that of the electric field is caused by the Coulomb force acting on the ions generated by charge injection near the electrode. The EHD pump is characterized by its simplicity, silent operation, compactness, and lightweight design. It has been theoretically and experimentally studied since the 1960s [Pickard, 1963a; Pickard, 1963b; Richter and Sandmaier, 1990], with research mainly centered around the MEMS field such as CPU cooling systems. Recently, it has gradually started to receive attention in the field of interaction as well [Morita et al., 2022; Morita et al., 2023; Smith et al., 2023].

Furthermore, an EHD pump can fundamentally be handmade, and we also have created one using copper as the electrode material. As shown in Figure 3, it consists of five layers: a base PET sheet, a comb-shaped copper tape electrode that triggers the EHD phenomenon, a VHB sheet acting as the flow path, an acrylic plate protecting the flow path, and a small tank housing the liquid. Long-term continuous operation, such as all day, can gradually wear the copper electrode, resulting in output variations. From an engineering perspective, these may be interpreted as difficult-to-control noises. However, considering a different perspective, producing

unpredictable, random movements has a high affinity for the concept of this work. Leveraging the characteristics of behaviors based on the EHD phenomenon, we attempt to create silent, beautiful movements filled with randomness in this work.

3 SILENT KINETIC MOVEMENT BY CONTROLLING THE CENTER OF GRAVITY OF THE LIQUID INSIDE

This work, as the name suggests, leverages the principles of a see-saw. Generally, the principle of a seesaw is that a force is applied to an effort point, creating a rotational movement around a pivot point. In this work, a pivot point is placed within the object and the center of gravity of the entire object corresponds to an effort point. A moderate amount of liquid is contained within the object, and by transporting this internal liquid, the center of gravity of the whole object is shifted. Figure 4 illustrates the specific operations. First, as shown in Figure 4 (i), a pivot point is set to serve as the axis of movement. Small tanks for holding the liquid are then arranged on both sides of the pivot point. The tanks are connected by an acrylic pipe, allowing the liquid to flow between them. Initially, as shown in Figure 4 (i), the liquid is housed in the left tank and the right tank is almost empty. Therefore, the overall center of gravity of the object is on the left, causing the left side to tilt downwards and the

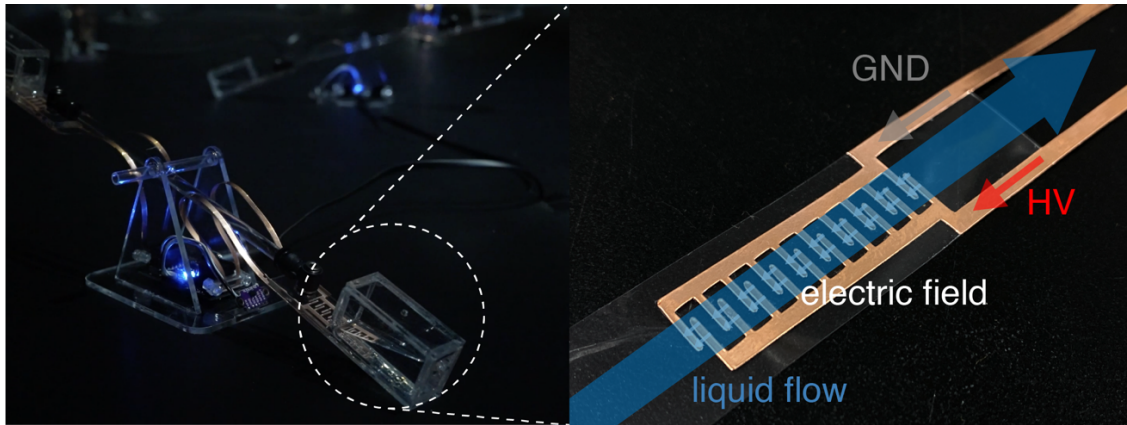


Figure 2: Mechanism of EHD phenomenon at the comb-shaped electrode. (©Takafumi Morita.)

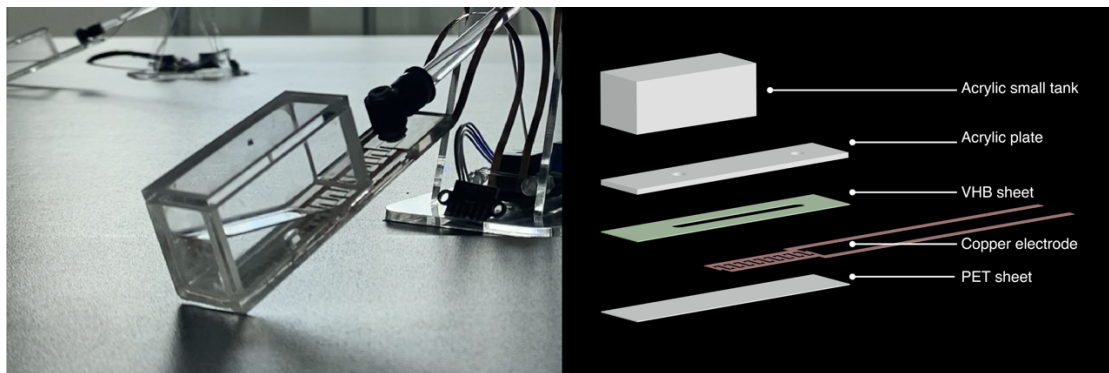


Figure 3: System diagram of the EHD pump unit embedded in the see-saw. (©Takafumi Morita.)

right side to tilt upwards. Then, as shown in Figure 4 (ii), by transporting the liquid from the left to the right, the center of gravity of the entire object switches from the left to the right. At this time, the pump on the left side is activated to generate liquid flow from left to right. As shown in Figure 4 (iii, iv), through the rotational movement around the pivot point, the left side rises and the right side descends. After this, the opposite series of operations causes the left side to descend and the right side to rise again. The see-saw repeats these movements to create a silent kinetic movement. In this way, by transporting the liquid internally, the center of gravity of the object changes, creating movement without mechanical actuators.

The system design of the see-saw is shown in Figure 5. The see-saw consists of three parts: the EHD pump that creates liquid flow, the electric part that controls the high voltage applied to the EHD pump, and the electrode that connects the EHD pump and the electric part while maintaining the electrical connection. The high voltage supplied from the electric part to the EHD pump uses the HVBT-5P-5 (Matsusada Precision Inc.). The HVBT-5P-5 is a cubic voltage amplifier of 12.7mm^3 that is controlled by the signal from the microcontroller. By applying a high voltage of 1 to 5kV, the EHD pump is driven. The Adafruit QT Py is used as a microcontroller, and its main processor is the ATSAMD21E18. This allows for the use of

a small and versatile Arduino development environment. The time-of-flight (TOF) sensor, VL53L0X, is used. This TOF sensor is placed on one side of the see-saw facing outwards. By detecting when the laser light pulse reflected from the surface of the target object returns and when it does not return, the microcontroller recognizes which way the seesaw is tilted. Based on this information, the microcontroller controls the appropriate EHD pump. The system basically operates in standalone mode. Although this work handles high voltage, the maximum output current is limited to the rated 0.1mA of the HVBT-5P-5, so there is no risk to human bodies.

4 AUDIENCE RECEPTIONS AT EXHIBITIONS

The see-saw has been exhibited twice at previous exhibitions. The first exhibition is shown in Figure 6, and the second exhibition is shown in Figure 7. As shown in Figures 7 and 8, A group of see-saw was arranged on a large desk for the first exhibition and on the floor for the second one. Each see-saw operates independently, and the overall movement never repeats the same pattern, varying according to the audience's angle and time. This randomness of movement provides a fresh experience for the audience. The silent and sparse movement of the see-saw allows the audience to feel a natural movement similar to the sway of tree leaves in the wind. As a result, while the exhibition space blends into the environment,

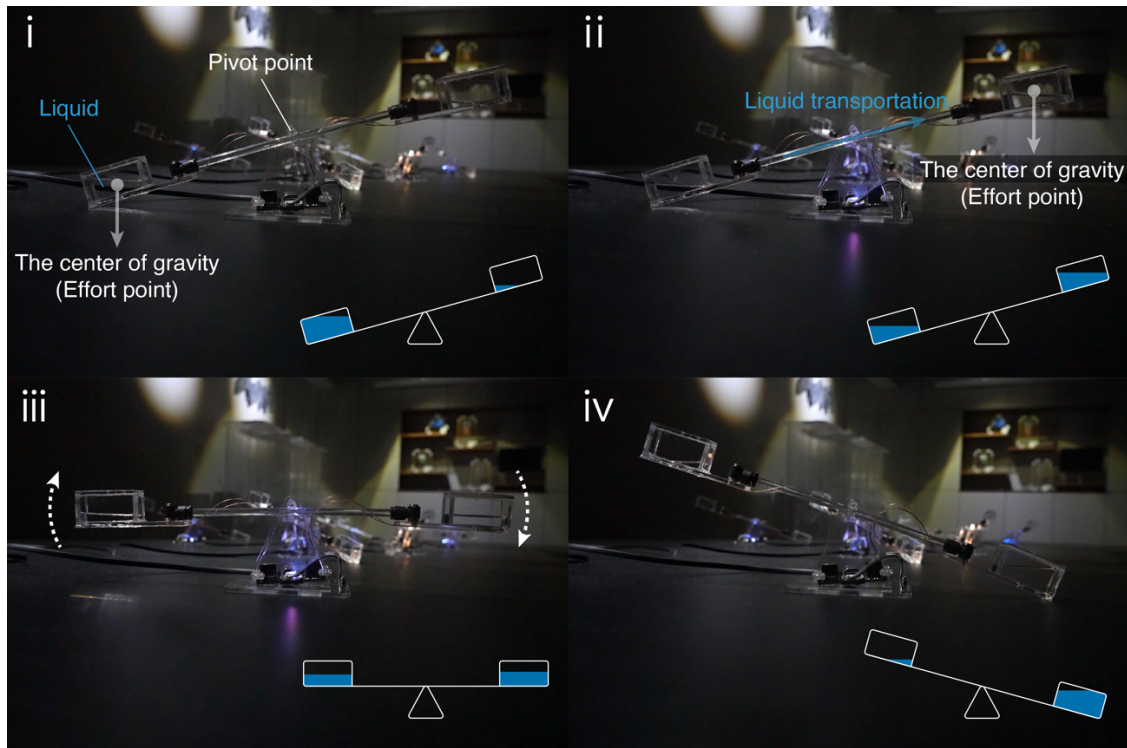


Figure 4: The basic operation of the see-saw. (i) The liquid is housed in the left tank and the right tank is almost empty, and the center of gravity of the object is on the left. (ii) The liquid is transported from the left to the right, and the center of gravity of the object switches from the left to the right. (iii) Through the rotational movement around the pivot point, the left side rises and the right side descends. (iv) The liquid is housed in the right tank and the left tank is almost empty, and the center of gravity of the object is on the right. (©Takafumi Morita.)

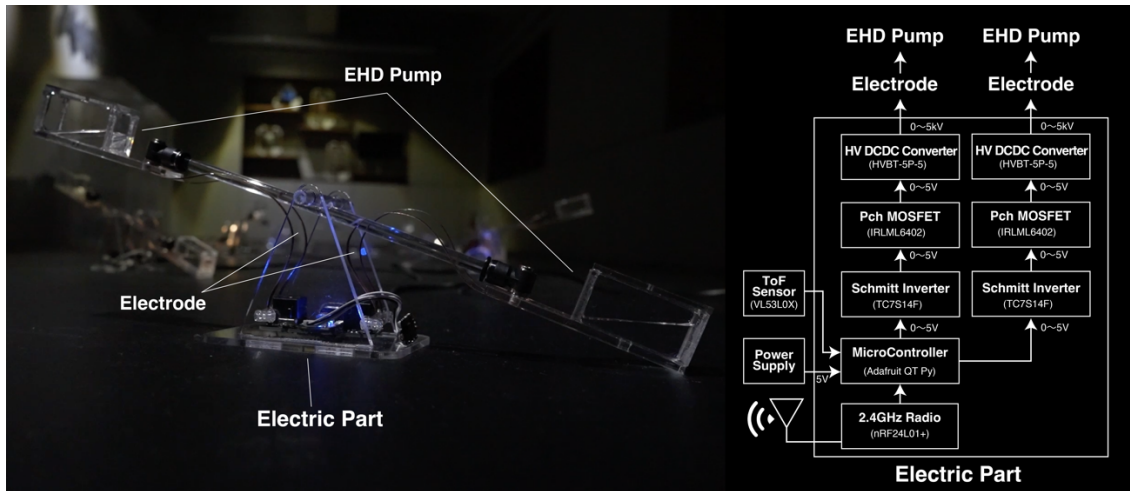


Figure 5: System design of the see-saw. The see-saw consists of the EHD pump, the electric part and the electrode. (©Takafumi Morita.)



Figure 6: Photos of the see-saw at Ars Electronica Festival 2022. (©Takafumi Morita. Photo: Yasuaki Kakehi.)

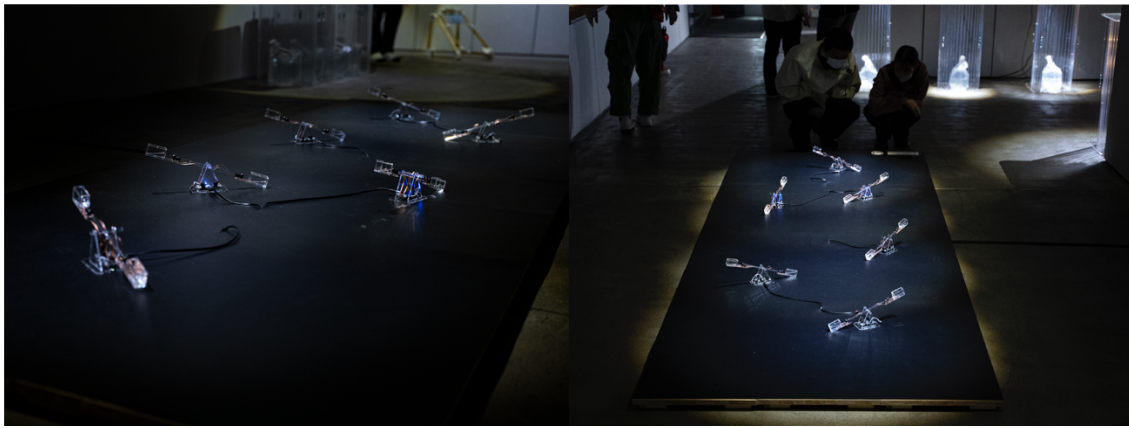


Figure 7: Photos of the see-saw at xlab showcase 2023. (©Takafumi Morita. Photo: Yasuaki Kakehi.)

it draws the attention of the audience as a unique object present there. Its vivid movement and silence create an intentional room, not to present all information, but to allow the audience to use their imagination and expand their inner interpretation. In this way, each audience member constructs their own unique imagination within their mind. This contributes to the creation of new artistic experiences, as emotions spring from this imaginative process.

Observing the reactions of the audience at the exhibition, it is difficult for them to grasp the whole picture of the see-saw at first glance because of its movement blending into the environment. However, most audiences stop to observe, captivated by the sparse movement of the see-saw. Many squat down and focus on the internal structure and operational principles of the see-saw. By consciously observing the see-saw, audiences understand that the tilt movement is generated by the change in the center of gravity of the entire object due to the liquid flow. Furthermore, most audiences were surprised by the sight of the liquid silently moving against gravity. We witnessed men and women of all ages observing our work with an intense curiosity about how it operates. One particularly striking aspect was the spontaneous occurrence of discussions among the audience in front of the see-saw, often accompanied by

gestures and expressions. We then felt that these reactions exactly matched our goal: to provide experiences that enable the audience to rediscover the beauty that is often overlooked in everyday life.

5 DISCUSSION

Many works have been created that utilize liquids. However, most of these works rely on mechanical pumps to move the liquid, generating significant operational noise. The see-saw proposed in this art paper is highly unique in that it produces no operational noise at all.

For the audience, the experience transcends the conventional expectation that objects in motion will produce sound. Normally, when objects move, noise is generated—whether it's the operation of a motor or the use of mechanical pumps to move water, operational sounds are always present. In contrast, this artwork aims to have the audience feel the movement of objects in silence. Furthermore, without using mechanical pumps, the artwork embeds sheet-like electric pumps at the bottom of tanks on both sides of see-saw, which seemingly lacks a visible driving source. These pumps allow the liquid to move against gravity, defying physical laws in a manner that appears to the audience. Such motion prompts the

audience to reconsider their general concept of how kinetic objects move.

Finally, we discuss the future potential for this artwork to be applied to the larger realm of technology-mediated experiences of the natural world. In this viewpoint, we think the necessity for silence becomes increasingly important. If objects can move without generating artificial operational noise, it's likely that animals and insects would be less disturbed. In such cases, the focus may not be limited to human interaction but could encompass interactions with the entire ecosystem, including animals and plants. From this perspective, we argue that artists should consider not only the pure creation but also the long-term effects of their work on the natural environment. This might involve using biodegradable materials, for instance. The future attitude of artists, as we believe, is to consider that the artwork can interact in a convivial and sustainable manner with the ecosystem without causing harm.

6 CONCLUSION

This work uses a silently operating electrohydrodynamic pump to move the liquid inside an object. This changes its center of gravity and expresses the tilting movement of the see-saw. The concept is to embody the margins of the movement, like plants swaying in the wind, which we typically overlook unless we pay conscious attention, to a physical object. We aimed to encourage the audience to re-recognize these potential movements that are often overlooked in everyday life, and hence we held two exhibitions of the see-saw. Taking into consideration the insights gained from our current work, in the future, we plan to continue exploring the possibilities of technology from a new perspective. This perspective includes considering margins and not dismissing elements with uncontrollable randomness as noise. Instead, we aim to respect their behavior based on their natural characteristics and mechanisms. By leveraging these qualities, we strive to create new artworks.

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